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THEORETICAL NEUROSCIENCE

TD6: SPIKE-TIMING-DEPENDENT PLASTICITY

All TD materials will be made available at `https://github.com/yfardella/Theo_Neuro_TD_26_27`.

In previous tutorials, we have developed models of point neurons responding to injection of currents. Biologically, these currents correspond to synaptic inputs from other neurons, making the synapse a key element of the communication between neurons. Importantly, synapses are thought to be the substrate of learning, as the modulation of their strength, or plasticity, modifies the interaction between neurons. Plasticity can obey different rules depending on the structure of synapses. In this tutorial, we propose to model long-term plasticity (through STDP).

1 Long term plasticity (LTP)

Spike-timing-dependent plasticity (STDP) is a class of long term modulation of synaptic efficacy, which depends on both pre-synaptic and post-synaptic mechanisms. In this sense, this is an associative process, which reinforces or weakens synaptic strengths between neurons based on their relative spike timings. This biological mechanism can be considered as an implementation of the Hebbian learning rule for adjusting the strength of connections between neurons. This computational rule has been proposed as a proxy for causality.

We consider a synapse connecting two neurons. The pre-synaptic neuron emits a train of n_{pre} pre-synaptic spikes $(t_{pre,1}, t_{pre,2}, \dots, t_{pre,n_{pre}})$, while the post-synaptic neuron emits a train of n_{post} post-synaptic spikes $(t_{post,1}, t_{post,2}, \dots, t_{post,n_{post}})$.

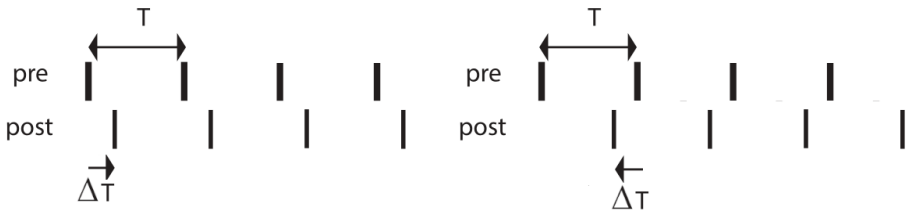
The synaptic weight w is modified according to the following rule: each pair of nearest-neighbour spikes $(t_{pre,i}, t_{post,j})$ induces a modification $\Delta w = f(t_{post,j} - t_{pre,i})$, with

$$f(s) = \begin{cases} A^+ e^{-s/\tau^+} \text{ if } s \geq 0, \\ -A^- e^{s/\tau^-} \text{ if } s < 0, \end{cases} \tag{1}$$

where A^+, A^-, τ^+ and τ^- are positive real parameters. Pairs of spikes are considered nearest-neighbour if and only if there is no other (pre-synaptic or post-synaptic) spike in the interval between the two spikes $]t_{pre,i}, t_{post,j}[$ (or $]t_{post,j}, t_{pre,i}[$). The synaptic modification is the sum of all modifications induced by the individual nearest-neighbour pairs.

1.1 Regular pre-synaptic train

Both neurons fire n spikes periodically with a frequency $F = 1/T$ (where T is the inter-spike interval), with a fixed interval ΔT between the spikes, $\Delta T = t_{post,i} - t_{pre,i}$ for all $i = 1, \ldots, n$.



1. Compute the total synaptic modification W induced by these spike trains for ΔT in $[-T/2, T/2]$.

   We have two cases to consider: either $\Delta T > 0$, either $\Delta T < 0$. The modification for one pair of spikes is as follows.

   For $\Delta T > 0$:

   - First, each pre-synaptic spike is followed by a post-synaptic spike ΔT later

   $$t_{post,j} - t_{pre,i} = \Delta T > 0 \Rightarrow \Delta w = A^+ e^{-\Delta T/\tau^+}.$$

   - Then, the post-synaptic spike is followed by a pre-synaptic spike $T - \Delta T$ later

   $$t_{post,j} - t_{pre,i} = -(t_{pre,i} - t_{post,j}) = -(T - \Delta T) < 0 \Rightarrow \Delta w = -A^- e^{(-T+\Delta T)/\tau^-}.$$

   For $\Delta T < 0$, the reasoning is strictly equivalent to considering $\Delta T > 0$ by simply interchanging the terms ΔT and $T - \Delta T$ in the previous equalities such that:

   - First, each pre-synaptic spike is followed by a post-synaptic spike $T - \Delta T$ later

   $$t_{post,j} - t_{pre,i} = T - \Delta T > 0 \Rightarrow \Delta w = A^+ e^{-(T-\Delta T)/\tau^+}.$$

   - Then, the post-synaptic spike is followed by a pre-synaptic spike ΔT later

   $$t_{post,j} - t_{pre,i} = -(t_{pre,i} - t_{post,j}) = -\Delta T < 0 \Rightarrow \Delta w = -A^- e^{\Delta T/\tau^-}.$$

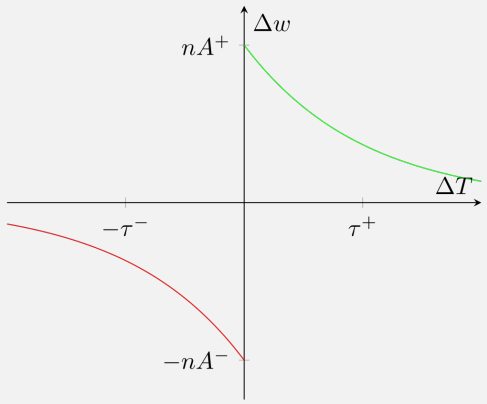
   The total synaptic modification after n pre and post-synaptic spikes is as follows.

   - For $\Delta T > 0 : \Delta w = n(A^+ e^{-\Delta T/\tau^+} - A^- e^{(-T+\Delta T)/\tau^-})$.
   - For $\Delta T < 0 : \Delta w = n(A^+ e^{-(T-\Delta T)/\tau^+} - A^- e^{\Delta T/\tau^-})$.

2. Compute the synaptic modification W when the inter-spike interval T is much longer than the widths of the STDP windows τ^+ and τ^-, for both $\Delta T > 0$ and $\Delta T < 0$. Sketch how W depends on ΔT.

Under this assumption, $e^{-T/\tau^{\pm}} \approx 0$ such that the expressions are simplified.

- For $\Delta T > 0$: $\Delta w \approx nA^{+}e^{-\Delta T/\tau^{+}}$.
- For $\Delta T < 0$: $\Delta w \approx -nA^{-}e^{\Delta T/\tau^{-}}$.



3. Compute the synaptic modification when the inter-spike interval T is much shorter than the widths of the STDP windows. Sketch again the dependence of W on ΔT.

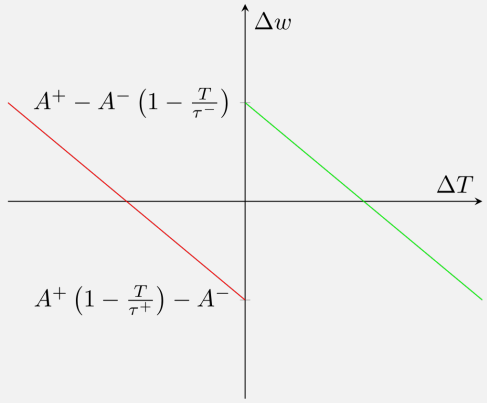
If $T \ll \tau^{+}, \tau^{-}$ then we also have that $\Delta T \ll \tau^{+}, \tau^{-}$. Under this assumption, the arguments of all the exponential terms tend to zero: we can therefore use the Taylor series expansion $e^{x} \approx 1 + x$.

- For $\Delta T > 0$:

$$\begin{aligned}\Delta w &\approx n\left(A^{+}(1-\Delta T/\tau^{+}) - A^{-}(1+(-T+\Delta T)/\tau^{-})\right)\\ &= n\left(A^{+} - A^{-}\left(1-\frac{T}{\tau^{-}}\right) - \Delta T\left(\frac{A^{+}}{\tau^{+}}+\frac{A^{-}}{\tau^{-}}\right)\right).\end{aligned}$$

- For $\Delta T < 0$:

$$\begin{aligned}\Delta w &\approx n\left(A^{+}(1-(T+\Delta T)/\tau^{+}) - A^{-}(1+\Delta T/\tau^{-})\right)\\ &= n\left(A^{+}\left(1-\frac{T}{\tau^{+}}\right) - A^{-} - \Delta T\left(\frac{A^{+}}{\tau^{+}}+\frac{A^{-}}{\tau^{-}}\right)\right).\end{aligned}$$



1.2 Stochastic pre-synaptic train

We now consider spikes produced randomly. For the sake of simplification, spikes still alternate between pre-synpatic and post-synaptic neurons, but the duration between successive spikes is stochastic. In total, $2n$ spikes are generated, n per neuron. Spikes are produced following a Poisson process of mean frequency $R = 1/T$ and assigned to each neuron in alternation.

4. Recalling that the probability density of inter-spike intervals is given by $p_{ISI}(t) = Re^{-Rt}$, what is the average total synaptic modification W at the end of the spike trains, as a function of n, R, A^+, A^-, τ^+ and τ^-?

   Reminder: let X be a continuous random variable. For any function $f : \mathbb{R} \to \mathbb{R}$,

   $$\mathbb{E}(f(X)) \overset{\text{def}}{=\!=} \int_{\mathbb{R}} f(x) p_X(x) dx,$$

   where p_X is the probability density function of X.

The expected synaptic modification after n spikes $\mathbb{E}(W)$ is the expectancy of a sum of $2n$ modifications Δw_i, because the spike train is composed of $2n$ inter-spikes intervals. By linearity of the expectancy,

$$\mathbb{E}(W) = \mathbb{E}\left(\sum_{i=1}^{2n} \Delta w_i\right) = \sum_{i=1}^{2n} \mathbb{E}(\Delta w_i).$$

Because of the hypothesis of alternation between the spikes of each neuron, those Δw_i are divided in two categories which follow their own probability law : n positive modifications (for intervals in which the pre-synaptic spike precedes the post-synaptic spike) and n negative modifications (for intervals in which the post-synaptic spike precedes the pre-synaptic spike).

Thus, the expectancy is proportional to the a single pre-post and post-pre couple of synaptic modifications $\mathbb{E}(\Delta w^+)$ and $\mathbb{E}(\Delta w^-)$ such that

$$\mathbb{E}(W) = n\mathbb{E}(\Delta w^+) + n\mathbb{E}(\Delta w^-) = n\left(\mathbb{E}(\Delta w^+) + \mathbb{E}(\Delta w^-)\right).$$

The expectancy is carried out over the ISI distribution (since $\Delta w^{\pm} = f(ISI)$)

$$\begin{aligned}
\mathbb{E}(W) &= n\left(\int_0^\infty A^+ e^{-t/\tau^+} p(t)dt - \int_0^\infty A^- e^{-t/\tau^-} p(t)dt\right) \\
&= n\left(A^+ \int_0^\infty e^{-t/\tau^+} Re^{-Rt} dt - A^- \int_0^\infty e^{-t/\tau^-} Re^{-Rt} dt\right) \\
&= nR\left(A^+ \int_0^\infty e^{-t(R+1/\tau^+)} dt - A^- \int_0^\infty e^{-t(R+1/\tau^-)} dt\right) \\
&= nR\left(A^+ \left[-\frac{e^{-t(R+1/\tau^+)}}{R+1/\tau^+}\right]_0^\infty - A^- \left[-\frac{e^{-t(R+1/\tau^-)}}{R+1/\tau^-}\right]_0^\infty\right) \\
&= nR\left(A^+ \frac{1}{R+1/\tau^+} - A^- \frac{1}{R+1/\tau^-}\right) = n\left(A^+ \frac{R\tau^+}{1+R\tau^+} - A^- \frac{R\tau^-}{1+R\tau^-}\right).
\end{aligned}$$

5. What are the limits of W in the low ($R \to 0$) and high ($R \to \infty$) frequency limits?

   - For $R \to 0$, $\mathbb{E}(W) \sim nR(A^+\tau^+ - A^-\tau^-)$.
   - For $R \to \infty$, $\mathbb{E}(W) \to n(A^+ - A^-)$.

6. In the BCM learning rule, the synaptic modification is negative at low frequencies, and positive at high frequencies. Determine the conditions on the parameters for implementing this rule.

   For W to be negative at low frequencies, we need

   $$nR(A^+\tau^+ - A^-\tau^-) < 0 \Leftrightarrow A^+\tau^+ < A^-\tau^- \Leftrightarrow \frac{A^+}{A^-} < \frac{\tau^-}{\tau^+}.$$

   For W to be positive at high frequencies, we need

   $$n(A^+ - A^-) > 0 \Leftrightarrow A^+ > A^- \Leftrightarrow 1 < \frac{A^+}{A^-}.$$

   For both conditions to hold, we therefore need

   $$1 < \frac{A^+}{A^-} < \frac{\tau^-}{\tau^+}.$$

7. Determine the frequency R_0 at which the total synaptic modification changes sign.

   The sign of the modification $\mathbb{E}(W)$ changes for R_0 such that

   $$\begin{aligned} \frac{A^+\tau^+}{1+R\tau^+} = \frac{A^-\tau^-}{1+R\tau^-} &\Rightarrow A^+\tau^+(1+R\tau^-) = A^-\tau^-(1+R\tau^+) \\ &\Rightarrow A^+\tau^+ - A^-\tau^- + R(A^+\tau^+\tau^- - A^-\tau^-\tau^+) = 0 \\ &\Rightarrow R_0 = \frac{A^-\tau^- - A^+\tau^+}{\tau^+\tau^-(A^+ - A^-)}. \end{aligned}$$